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Except where otherwise indicated, this thesis is my own original work.

Mingyang Sheng
29 July 2026

to my xxx, yyy (yyy is the people you want to dedicated this thesis to.)

Acknowledgments

Who do you want to thank?

Abstract

Put your abstract here.

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Introduction

In many practical statistical applications, data are often incomplete due to missing observations, measurement corruption, or the presence of latent variables. These issues make the computation of maximum likelihood estimate (MLE) considerably more difficult than in fully observed settings. The Expectation–Maximization (EM) algorithm was developed to address this challenge, and its effectiveness has been further enhanced by advances in modern computing. Consequently, EM has become a fundamental method for likelihood-based inference with incomplete data, leading to an extensive body of research on its computational and statistical properties.

However, their use in high-dimensional applications may be very limited as both the storage requirements and the additional computational cost per-iteration can quickly become burdensome as the number of model parameters grows.

Comparing Existing Algorithms & Our Work

Research on Our EM Algorithm

Organization of Thesis How many chapters you have? You may have Chapter 2, Chapter 3, Chapter 4, Chapter 7, and Chapter 8.

Literature Review

High-dimensional multivariate count data with group structure arises across many modern applications. In computational linguistics, documents partition into authors or topics, and each document is a vocabulary-sized word-count vector [Taddy, 2015]. In micro-biome ecology, samples nest within subjects while the response be a vector of taxonomic abundances over hundreds of taxa [Tu and Subedi, 2023]. Despite their diversity, these settings share a common structure with I observations in J groups, whose individual observation is a count vector $\mathbf{y}_{ij} \in \mathbb{Z}_{\geq 0}^Q$ over Q categories.

The modeling challenge has two dimensions. The compositional aspect calls for the Multinomial distribution with a log-linear softmax link. The hierarchical aspect requires group-level random effects to capture dependence among observations from the same group. Ignoring this structure inflates Type I error and introduces confounding bias in fixed-effect estimates [Lee et al., 2019]. The literature has addressed these two dimensions largely in isolation. The present work provides a synthesis suited to the large- Q regime.

The simplest approach treats observations as exchangeable and attributes all variation to measured covariates via Multinomial logistic regression. For large Q , direct Multinomial maximum likelihood is computationally prohibitive owing to the coupled softmax normalizer. Baker [1994] established the Poisson–Multinomial Equivalence (PME). Conditional on the total count M_i , the Multinomial likelihood factors can be separated into Q independent Poisson log-linear models with a common additive offset ξ_{ij} . Taddy [2015] exploited this to propose Distributed Multinomial Regression (DMR) and applied it to large-vocabulary document classification. Lee et al. [2017] subsequently formalized and extended the PME framework clarifying conditions under which it applies to a broad class of Multinomial regression models.

DMR and standard MLR remain important baselines, but their central limitation is the absence of group-level random effects. When groups differ systematically, treating observations as exchangeable absorbs unmodeled group variation into fixed-effect estimates. For the large- Q setting with high-dimensional fixed effects, Correia et al. [2020] provides efficient Poisson estimation algorithms. Our work builds on this computational tradition while additionally modeling group-level structure.

Aitchison and Ho [1989] introduced the Poisson Lognormal (PLN) distribution placing a Gaussian prior $\mathbf{Z}_i \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ on the log-linear predictors and drawing $Y_{ijq} \mid \mathbf{Z}_{iq} \sim \text{Poisson}(\exp(\mathbf{Z}_{iq}))$ independently. PLN captures between-category cor-

relations and over-dispersion. However, they can not estimate a full $Q \times Q$ covariance Σ in large Q regime.

Chiquet et al. [2018] addressed this via PLN-PCA, which constraining $\Sigma = \mathbf{C}\mathbf{C}^\top$ with $\mathbf{C} \in \mathbb{R}^{Q \times K}$ and $K \ll Q$, then used variational EM with a mean-field Gaussian approximation. PLN-PCA imposes a low-rank factor covariance, who has the closest structural antecedent to our model. The key distinctions are that PLN-PCA operates at the observation level only and no group random effect. Also, they provided no large- Q asymptotic theory.

Tu and Subedi [2023] introduced logistic-normal multinomial (LNM) factor analyzers with a factor-structured observation-level covariance $\Sigma = \mathbf{B}\mathbf{B}^\top + \Psi$, $\Psi = \text{diag}(\psi_1, \dots, \psi_Q)$. The EM algorithm uses a variational Gaussian approximation in the E-step and the Rubin-Thayer iterations [Rubin and Thayer, 1982] in the M-step. Their LNM-FA shares the factor-analytic motivation of the present work, which point motivated our work targeting to structure a group-level random effect $\lambda_j = \mathbf{B}\mathbf{f}_j + \epsilon_j$, adopting the spherical simplification $\mathbf{D} = \sigma^2 \mathbf{I}_Q$.

Relative to PLN-PCA, we add a group-level hierarchical structure, a spherical idiosyncratic term, and Laplace inference. Relative to LNM-FA, the principal advances are the group-level factor structure and the large- Q consistency theory. Relative to DMR, we add group-level random effects with a factor prior, thus DMR can be recovered as the special case when $J = 1$.

Model Setup & Identifiability Conditions

Given the notation of our thesis in section 3.1, we begin with basic background of the Poisson-Multinomial Equivalence in section 3.2 and EM (Expectation-Maximization) algorithm in section 3.3, which are the core tricks of our algorithms in Chapter 4 and Chapter 5. Our model structure is provided in section 3.4. In section 3.5, we give the identifiable condition for loading matrix so that the it can be recovered uniquely.

3.1 Model Setup

Our theoretical data structure adapt the yogurt data used by Lee et al. [2017]. Let $\mathbf{y}_{ij} = (y_{ij1}, \dots, y_{ijQ})^\top \in \mathbb{Z}_{\geq 0}^Q$ denote the count vector representing category-specific choices for observation i in group $j(i)$, spanning Q distinct categories. For each observation i , let $j(i) \in \{1, 2, \dots, J\}$ maps the individual to its corresponding cluster group. We define the group-specific index set as $\mathcal{I}_j = \{i : j(i) = j\}$ with its cardinality denoted by $I_j = |\mathcal{I}_j|$ such that the aggregate sample size satisfies $I = \sum_{j=1}^J I_j$. Here y_{ij} denotes the Multinomial sum for observation i in group $j(i)$ across Q categories. Each observation carries a covariate vector $\mathbf{v}_{ij} \in \mathbb{R}^P$. We omit a constant term from $\mathbf{v}_{ij}$ to avoid redundancy with the category-specific intercepts already captured by $\boldsymbol{\mu}$.

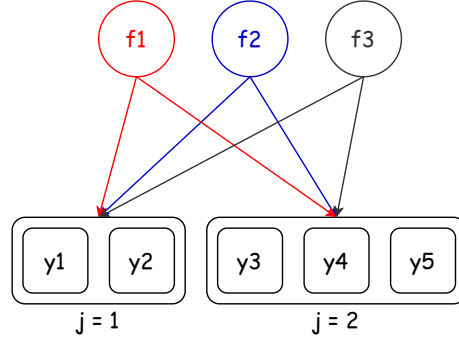


Figure 3.1: Data Structure: Orthogonal case that each latent factors $f1$ – $f3$ is independent to others. Each directed edge represents that a latent factor affects the response variable y_{ij} for $j \in \{1, 2\}$. Notice that each latent factor affects all groups publicly.

3.2 Poisson-Multinomial Equivalence

The Poisson-Multinomial Equivalence established by Baker [1994] states that the Multinomial likelihood conditional on $y_{ij\cdot}$ is exactly equivalent to a product of independent Poisson likelihoods sharing a common additive offset,

$$\xi_{ij} = \log y_{ij\cdot} - \log \sum_{q=1}^Q \exp(\eta_{ijq}), \quad (3.1)$$

The joint Multinomial draw $\mathbf{y}_{ij} \mid \lambda_j \sim \text{Multinomial}(y_{ij\cdot}, \mathbf{p}_{ij})$ is equivalent to $y_{ijq} \mid \lambda_j \sim \text{Poisson}(\exp(\eta_{ijq} + \xi_{ij}))$ independently for $q = 1, \dots, Q$, conditional on $y_{ij\cdot}$. We give the definition of η_{ijq} and $\mathbf{p}_{ij}$ in section 3.4. The offset (3.1) absorbs the softmax normalizer and decouples the Q Poisson problems, thus we can solve these Q Poisson independently.

Lee et al. [2017] extended the framework of Baker [1994] by treating the aggregate count $y_{ij\cdot}$ as Poisson-distributed given the group effect λ_j , rather than fixing it as the conditioning variable in the Multinomial likelihood. Marginalizing over this random aggregate recovers the same factorization into independent Poisson components across categories $q = 1, \dots, Q$. As a result, ξ_{ij} becomes the rate parameter of this induced Poisson distribution instead of a fixed conditioning quantity.

This chapter carries the same idea one step further into E-step. In the standard Laplace E-step, λ_j is updated via Newton’s method (4.7) on the full Multinomial log-posterior $\ell(\lambda_j)$. The bottleneck is the term in Hessian $\sum_{i \in \mathcal{I}_j} y_{ij\cdot} \mathbf{p}_{ij} \mathbf{p}_{ij}^\top$, which comes from the softmax normalizer with rank- I_j . This term couples all Q categories together, so each Newton step requires an $O(Q^3)$ linear solve. Inspired by the insight of Lee et al. [2017], we introduce ξ_{ij} as an explicit auxiliary variable in the E-step. Holding ξ_{ij} fixed while updating λ_j removes the log-sum-exp coupling entirely, so the Hessian reduces to a diagonal matrix, enabling a second Woodbury reduction cutting the per-step cost from $O(Q(I_j + K))$ down to $O(QK^2 + K^3)$.

3.3 EM Algorithm & Relatives

In order to have a good understanding for further discussion, we briefly review the classical EM algorithm in this section. Then we discuss the models that have the similar structure with our model.

Let Y and Z be random variables taking values in the sample spaces $\mathcal{Y}$ and $\mathcal{Z}$, respectively. Suppose that the pair (Y, Z) has a joint density function f_{θ^*} that belongs to some parameterized family $\{f_{\theta} \mid \theta \in \Omega\}$, for a non-empty compact convex set Ω . Rather than observing complete data (Y, Z) , we observe only component Y . Thus, the component Z corresponds to the missing or latent structure in the data. Our goal is to obtain an estimate of the unknown parameter θ^* via maximum likelihood to compute some $\hat{\theta} \in \Omega$ maximizing the function $\theta \mapsto g_{\theta}(y)$, where

$$g_{\theta}(y) = \int_{\mathcal{Z}} f_{\theta}(y, z) dz \quad (3.2)$$

is the density function of the observed variable Y .

In many settings, computing log likelihood $\log f_{\theta}(y, z)$ of both latent variables and observed data is easier and computationally cheaper than evaluating the log likelihood of the observed data itself. The EM algorithm is well-suited to such settings. For each $\theta \in \Omega$, let $k_{\theta}(z \mid y)$ denote the conditional density of z given by y , then the log likelihood at $\theta' \in \Omega$ can be lower bounded using Jensen's inequality

$$\log g_{\theta'}(y) \geq \underbrace{\int_{\mathcal{Z}} k_{\theta'}(z \mid y) \log f_{\theta'}(y, z) dz}_{Q(\theta' \mid \theta)} - \int_{\mathcal{Z}} k_{\theta'}(z \mid y) \log k_{\theta}(z \mid y) dz, \quad (3.3)$$

the equality holds when $\theta = \theta'$. Thus, we have a family of lower bounds on the log likelihood, denoted by $Q(\cdot \mid \cdot)$. We maximize this lower bound in M-step

$$\theta^{(t+1)} = \arg \max_{\theta' \in \Omega} Q(\theta' \mid \theta^{(t)}), \quad (3.4)$$

and then re-evaluates the lower bound at the new parameter value in E-step.

3.4 Two-Level Hierarchy

Observation Level Conditional on the total count $y_{ij\cdot} = \sum_{q=1}^Q y_{ijq}$ and group-level random effect $\lambda_j \in \mathbb{R}^Q$, each observation follows a Multinomial distribution:

$$\mathbf{y}_{ij} \mid y_{ij\cdot}, \lambda_j \sim \text{Multinomial}(y_{ij\cdot}, \mathbf{p}_{ij}), \quad p_{ijq} = \frac{\exp(\eta_{ijq})}{\sum_{q'=1}^Q \exp(\eta_{ijq'})}, \quad (3.5)$$

where the log-linear predictor is:

$$\eta_{ijq} = \mu_q + \mathbf{v}_{ij}^{\top} \boldsymbol{\phi}_q + \lambda_{jq}, \quad q = 1, 2, \dots, Q, \quad (3.6)$$

and $\mu_q \in \mathbb{R}$ is the category-specific intercept, $\boldsymbol{\phi}_q \in \mathbb{R}^P$ is the vector of covariate effects for category q , and λ_{jq} is the contribution of group j to the log-relative-rate of category q . Equation (3.6) extends the linear predictor of DMR [Taddy, 2015] and of the Poisson extension of Lee et al. [2017]; the three models differ only in the distribution placed on λ_j .

The softmax link in (3.5) is invariant under the shift $\eta_{ijq} \mapsto \eta_{ijq} + c_{ij}$ for any constant c_{ij} common to all categories, so the intercepts $\boldsymbol{\mu} = (\mu_1, \dots, \mu_Q)$ are identified only up to an additive constant. We remove the indeterminacy by imposing the reference constraint $\mu_1 = 0$ (equivalently a sum-to-zero constraint $\sum_q \mu_q = 0$). This indeterminacy restores automatically by the Poisson representation introduced below, in which each category carries its own offset.

Throughout, we treat the total count y_{ij} as ancillary for $(\boldsymbol{\mu}, \boldsymbol{\Phi}, \boldsymbol{\lambda}_j)$. We assume its generating mechanism carries no information about these parameters beyond what they already encoded, so conditioning on y_{ij} in (3.5) loses no relevant information. This is standard in Multinomial regression, but it is not automatic. We do not attempt a joint model of volume and composition here. To be clear, ancillarity is about information, not determinism: the total is still random, its distribution just carries no extra information about $(\boldsymbol{\mu}, \boldsymbol{\Phi}, \boldsymbol{\lambda}_j)$. In Section 3.2, we use the Poisson-Multinomial equivalence to represent each y_{ijq} as an independent Poisson variate, whose offset ξ_{ij} absorbs the (random) total. Conditioning on y_{ij} recovers the Multinomial conditional in (3.5) exactly. We treat ξ_{ij} as a computational device but not a modeling claim about the total.

Group Level The group random effects are assumed independently and identically distributed:

$$\boldsymbol{\lambda}_j \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \quad \boldsymbol{\Sigma} = \mathbf{B}\mathbf{B}^\top + \mathbf{D}, \quad (3.7)$$

where $\mathbf{B} \in \mathbb{R}^{Q \times K}$ is the factor loadings matrix and $\mathbf{D}$ is the variance. Equivalently, in the factor regression form:

$$\boldsymbol{\lambda}_j = \mathbf{B}\mathbf{f}_j + \boldsymbol{\epsilon}_j, \quad \mathbf{f}_j \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_K), \quad \boldsymbol{\epsilon}_j \sim \mathcal{N}(\mathbf{0}, \mathbf{D}), \quad (3.8)$$

so that $\lambda_{jq} = \mathbf{b}_q^\top \mathbf{f}_j + \epsilon_{jq}$, where $\mathbf{b}_q$ is the q -th row of $\mathbf{B}$. We use (3.7) for inference and (3.8) for interpretation.

We assume $\mathbf{D} = \sigma^2 \mathbf{I}_Q$ with $\sigma^2 > 0$ rather than the fully diagonal $\mathbf{D} = \text{diag}(d_1^2, \dots, d_Q^2)$ with entries $d_q > 0$. This reflects the modeling view that, once the factor structure $\mathbf{B}$ and the intercepts $\boldsymbol{\mu}$ have absorbed the systematic between-category structure, the residual noise has no a priori reason to differ across categories. It also keeps the parameter count in $\mathbf{D}$ from growing with Q . As a further benefit, this choice simplifies the Woodbury reduction introduced in Chapter 5, whose complexity expression is considerably harder to state under the fully diagonal assumption.

3.5 Identifiability

In many settings, we categorize the latent factors into orthogonal and non-orthogonal [Cui and Xu, 2026]. Orthogonal factors are assumed to be independent, which illustrates the relationship of factors in our group random effects (3.8). In this section, we present the conditions that guarantee the rotational identifiability, which are treated as given in the rest of the thesis.

Both linear latent factor model and the generalized latent factor model suffer from rotational indeterminacy, which is the primary source of non-identification in factor analysis (Bai and Li [2012], Bai and Ng [2013]). The factor structure (3.8) is subject to classical indeterminacy, which includes both rotational and sign indeterminacy. For any $K \times K$ orthogonal matrix $\mathbf{V}$, the substitution $(\mathbf{B}, \mathbf{f}_j) \rightarrow (\mathbf{B}\mathbf{V}, \mathbf{V}^\top \mathbf{f}_j)$ leaves the distribution of λ_j unchanged. Also, the substitution $(\mathbf{b}_k, \mathbf{f}_j) \rightarrow (-\mathbf{b}_k, -\mathbf{f}_j)$ product of $\mathbf{b}_q \mathbf{f}_k$ is the same as $(-\mathbf{b}_q)(-\mathbf{f}_k)$.

A more common approach in applications to ensure identifiability is to prescribe zeros in the loading matrix $\mathbf{B}$. Under our orthogonal assumption, the additional $K(K-1)/2$ constraints are imposed by fixing $K(K-1)/2$ entries to be zeros. Interestingly, the Example (1) shows that not all configurations of the $K(K-1)/2$ zeros ensure identifiability.

Example 1. This example was given by Cui and Xu [2026]. We set $K = 3$ and assign at least 3 zeros in loading matrix $\mathbf{B}$. Consider the loading matrix $\mathbf{B}$ and the orthogonal transformation matrix $\mathbf{G}$,

$$\mathbf{B}_{[1:2]} = \begin{pmatrix} 0 & a_{12} & a_{13} \\ a_{21} & 0 & 0 \end{pmatrix}, \quad \mathbf{G}_\theta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix}.$$

For any θ ,

$$[\mathbf{B}\mathbf{G}_\theta]_{[1:2]} = \begin{pmatrix} 0 & *_{12} & *_{13} \\ a_{21} & 0 & 0 \end{pmatrix}.$$

For any θ , the zero pattern of $[\mathbf{B}\mathbf{G}_\theta]_{[1:2]}$ retains the same as $\mathbf{B}_{[1:2]}$. Thus, the above transformation matrix $\mathbf{G}_\theta$ leaves the factor covariance matrix unchanged, which is not rotationally identifiable under this constraint.

Example (1) shows that a necessary condition for identifiability is that no two columns of the loading matrix share the same zero pattern. To avoid these cases, we consider the following sufficient condition that covers many commonly used identifiability constraints. Thus, we give the positive lower triangular constraint to recover the true value of matrix $\mathbf{B}$,

$$b_{qk} = 0 \text{ for } q < k, \quad b_{kk} > 0 \text{ for } k = 1, 2, \dots, K, \quad (3.9)$$

which solves the rotationally indeterminacy and sign indeterminacy. Cui and Xu [2026] had proved many sufficient constraints, and we adapt their IC⁽²⁾ with orthogonal latent variables in factor analysis.

Methodology

The observed-data log-likelihood under the model of Chapter 3 is

$$\ell(\boldsymbol{\theta}) = \sum_{j=1}^J \sum_{i \in \mathcal{I}_j} \log p(\mathbf{y}_{ij} \mid \boldsymbol{\theta}) = \sum_{j=1}^J \sum_{i \in \mathcal{I}_j} \log \int p(\mathbf{y}_{ij} \mid \boldsymbol{\lambda}_j) p(\boldsymbol{\lambda}_j \mid \boldsymbol{\Sigma}) d\boldsymbol{\lambda}_j, \quad (4.1)$$

where the parameter space is $\boldsymbol{\theta} := \{\boldsymbol{\mu}, \boldsymbol{\Phi}, \mathbf{B}, \sigma^2\}$. The integral in (4.1) has no closed form, because the Gaussian prior on $\boldsymbol{\lambda}_j$ is not conjugate to the multinomial likelihood, so we treat $\{\boldsymbol{\lambda}_j\}_{j=1}^J$ as missing data and then estimate $\boldsymbol{\theta} = \{\boldsymbol{\mu}, \boldsymbol{\Phi}, \mathbf{B}, \sigma^2\}$ by EM. The standard approach in generalized linear mixed models is the Laplace approximation ([Breslow and Clayton, 1993], [Tierney and Kadane, 1986]), which replaced the integral with a Gaussian approximation at the posterior mode.

4.1 The Choice of Approximation Level

A natural approach motivated by Chiquet et al. [2018] and their PLN-PCA model approximates the posterior of the K -dimensional factor $p(\mathbf{f}_j \mid \mathbf{y}_{ij})$. Since we assumed $K \ll Q$, this reduces the Newton-step Hessian from $Q \times Q$ to $K \times K$, which appears computationally attractive. However, this approach encounters two fundamental obstacles.

Group Collapsing Within group j , observations $\{i \in \mathcal{I}_j\}$ share the same factor $\mathbf{f}_j$ but carry distinct covariate vectors $\mathbf{v}_{ij}$ and total count y_{ij} . The factor-space log-posterior requires summing $\sum_{i \in \mathcal{I}_j} \log p(\mathbf{y}_{ij} \mid \mathbf{f}_j)$, where each term involves its own normalizing constant $\sum_q \exp(\eta_{ijq}(\mathbf{f}_j))$. A naive implementation collapses all observations in group j into a single super-observation with total counts $\mathbf{y}_{\cdot j} = \sum_{i \in \mathcal{I}_j} \mathbf{y}_{ij}$ and averaged fixed effects $\bar{\boldsymbol{\eta}}_j$, treating the group as a single Multinomial draw. This is mathematically incorrect. The Multinomial log-normalizer does not linearize under summation, and the approximation error grows with within-group heterogeneity in $\mathbf{v}_{ij}$.

Posterior Second-Moment Deflation The factor-space Laplace posterior yields a second-moment matrix $\mathbf{W}_f = J^{-1} \sum_j (\hat{\mathbf{f}}_j \hat{\mathbf{f}}_j^\top + \hat{\mathbf{V}}_j)$ that systematically underestimates

$\mathbf{I}_K$ due to Laplace shrinkage. Because the M-step for $\mathbf{B}$ receives only $\mathbf{B}\mathbf{W}_f\mathbf{B}^\top$ as its sufficient statistic for $\mathbf{\Sigma}$, the Rubin–Thayer algorithm (Rubin and Thayer [1982]) compensates by inflating $\hat{\mathbf{B}}$, leading to the error in estimation. Thus, instead of approximate the posterior of K dimensional latent factors, we do the approximation on Q dimensional group random effect $p(\lambda_j | \mathbf{y}_{ij})$ directly and move the factor decomposition from the E-step to the M-step. The E-step treats λ_j as an unknown Q -vector with Gaussian prior $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$, then the M-step decomposes the resulting estimated covariance $\hat{\mathbf{\Sigma}}_{\text{obs}}$ into $\mathbf{B}\mathbf{B}^\top + \sigma^2\mathbf{I}$.

4.2 E-Step: Laplace Approximation

For each group j , the log-posterior of λ_j is given by:

$$\mathcal{L}(\lambda_j) = \sum_{i \in \mathcal{I}_j} \left[\sum_q y_{ijq} \eta_{ijq}(\lambda_j) - y_{ij} \cdot \log \sum_q \exp(\eta_{ijq}(\lambda_j)) \right] - \frac{1}{2} \lambda_j^\top \mathbf{\Sigma}^{-1} \lambda_j, \quad (4.2)$$

up to an additive constant independent of λ_j , where

$$\eta_{ijq}(\lambda_j) = \mu_q + \mathbf{v}_{ij}^\top \boldsymbol{\phi}_q + \lambda_{jq}, \quad (4.3)$$

is evaluated separately for each $i \in \mathcal{I}_j$, gives the individual-specific covariate structure. The gradient and Hessian of (4.2) are

$$\nabla_{\lambda_j} \ell = \sum_{i \in \mathcal{I}_j} (\mathbf{y}_{ij} - y_{ij} \cdot \mathbf{p}_{ij}(\lambda_j)) - \mathbf{\Sigma}^{-1} \lambda_j, \quad (4.4)$$

$$\mathbf{H}(\lambda_j) = - \sum_{i \in \mathcal{I}_j} y_{ij} \cdot \left(\text{diag}(\mathbf{p}_{ij}) - \mathbf{p}_{ij} \mathbf{p}_{ij}^\top \right) - \mathbf{\Sigma}^{-1}, \quad (4.5)$$

where $\mathbf{p}_{ij}(\lambda_j) = \text{softmax}(\boldsymbol{\eta}_{ij}(\lambda_j))$. Since $\mathbf{P}_{ij} \succeq 0$ and $\mathbf{\Sigma}^{-1} \succ 0$, we have $\mathbf{H}(\lambda_j) \prec 0$ everywhere, so $\ell(\lambda_j)$ is strictly concave and has a unique maximizer

$$\hat{\lambda}_j = \arg \max_{\lambda \in \mathbb{R}^Q} \ell_j(\lambda), \quad (4.6)$$

which can be found by Newton’s method with Armijo backtracking search with step size $s^{(t)} \in (0, 1]$. Strict concavity guarantees convergence to $\hat{\lambda}_j$ from any starting point. The Laplace approximation then replaces the intractable posterior with the Gaussian matched to the model and curvature at,

$$\lambda_j^{(t+1)} = \lambda_j^{(t)} - s^{(t)} \mathbf{H}(\lambda_j^{(t)})^{-1} \nabla_{\lambda_j} \ell(\lambda_j^{(t)}), \quad (4.7)$$

with backtracking step size $s^{(t)}$ converges to the unique global maximum $\hat{\lambda}_j$ from any initialization. The Laplace approximation then yields

$$p(\lambda_j | \mathbf{y}_{ij}) \approx \mathcal{N}(\hat{\lambda}_j, \hat{\mathbf{S}}_j), \quad \hat{\mathbf{S}}_j = (-\mathbf{H}(\hat{\lambda}_j))^{-1}. \quad (4.8)$$

Each Newton step requires forming and solving the $Q \times Q$ system in (4.5) at a cost of $\mathcal{O}(I_j Q^2 + Q^3)$ per step. We will discuss and remove this bottleneck in Chapter 5.

4.3 M-Step (a): Fixed Effects via Poisson GLMs

With $\hat{\lambda}_j$ fixed, recompute the Poisson-Multinomial Equivalence offset:

$$\xi_{ij} = \log y_{ij} - \log \sum_q \exp(\hat{\mu}_q + \mathbf{v}_{ij}^\top \hat{\boldsymbol{\phi}}_q + \hat{\lambda}_{jq}). \quad (4.9)$$

For $q = 1, 2, \dots, Q$, we solve the maximization problem independently:

$$(\hat{\mu}_q, \hat{\boldsymbol{\phi}}_q) = \arg \max_{\mu_q, \boldsymbol{\phi}_q} \sum_i [y_{ijq}(\mu_q + \mathbf{v}_{ij}^\top \boldsymbol{\phi}_q) - \exp(\mu_q + \mathbf{v}_{ij}^\top \boldsymbol{\phi}_q + \hat{\lambda}_{jq} + \xi_{ij})]. \quad (4.10)$$

This is a Poisson log-linear regression with offset $\lambda_{jq} + \xi_{ij}$, solved by iteratively reweighted least squares. The Q problems are independent and may be parallelized.

4.4 M-Step (b): Factor Loadings and Idiosyncratic Variance

The EM sufficient statistic for $\boldsymbol{\Sigma}$ follows directly from the Laplace output:

$$\hat{\boldsymbol{\Sigma}}_{\text{obs}} = \frac{1}{J} \sum_j \mathbb{E}[\boldsymbol{\lambda}_j \boldsymbol{\lambda}_j^\top \mid \mathbf{y}_{ij}] \approx \frac{1}{J} \sum_j (\hat{\boldsymbol{\lambda}}_j \hat{\boldsymbol{\lambda}}_j^\top + \hat{\mathbf{S}}_j). \quad (4.11)$$

Critically, $\hat{\boldsymbol{\Sigma}}_{\text{obs}}$ is fully determined by current Laplace output $(\hat{\boldsymbol{\lambda}}_j, \hat{\mathbf{S}}_j)$ and does not need to be re-evaluated during the Rubin-Thayer sub-iterations, thus eliminating the circular dependency inherent in factor-space formulations. Given $\hat{\boldsymbol{\Sigma}}_{\text{obs}}$, we decompose $\boldsymbol{\Sigma} = \mathbf{B}\mathbf{B}^\top + \sigma^2 \mathbf{I}_Q$ via the Rubin-Thayer EM factor analysis algorithm (Rubin and Thayer [1982])

$$\hat{\boldsymbol{\beta}} = \hat{\mathbf{B}}^\top (\hat{\mathbf{B}}\hat{\mathbf{B}}^\top + \hat{\sigma}^2 \mathbf{I}_Q)^{-1}, \quad (4.12)$$

$$\hat{\boldsymbol{\Theta}} = \mathbf{I}_K - \hat{\boldsymbol{\beta}}\hat{\mathbf{B}} + \hat{\boldsymbol{\beta}}\hat{\boldsymbol{\Sigma}}_{\text{obs}}\hat{\boldsymbol{\beta}}^\top, \quad (4.13)$$

$$\hat{\mathbf{B}}_{\text{new}} = \hat{\boldsymbol{\Sigma}}_{\text{obs}}\hat{\boldsymbol{\beta}}^\top \hat{\boldsymbol{\Theta}}^{-1}, \quad (4.14)$$

$$\hat{\sigma}_{\text{new}}^2 = \frac{1}{Q} \text{tr}(\hat{\boldsymbol{\Sigma}}_{\text{obs}} - \hat{\mathbf{B}}_{\text{new}}\hat{\boldsymbol{\beta}}\hat{\boldsymbol{\Sigma}}_{\text{obs}}). \quad (4.15)$$

The iterations (4.12)-(4.15) are repeated to convergence. The PLT constraint (3.9) is enforced post-hoc via a QR decomposition of the top K rows of $\hat{\mathbf{B}}_{\text{new}}$, followed by sign correction to ensure positive diagonal entries and zeroing of the upper triangle. Applying PLT only after the Rubin-Thayer sub-iterations converge, rather than at each inner step, avoids the discontinuities that would otherwise disrupt convergence. Given $\hat{\mathbf{B}}$ and $\hat{\sigma}^2$, the factor scores are recovered by the Bayes linear estimator [Gelman

et al., 2013]:

$$\hat{\mathbf{f}}_j = \left(\mathbf{I}_K + \frac{1}{\hat{\sigma}^2} \hat{\mathbf{B}}^\top \hat{\mathbf{B}} \right)^{-1} \frac{1}{\hat{\sigma}^2} \hat{\mathbf{B}}^\top \hat{\boldsymbol{\lambda}}_j, \quad (4.16)$$

requiring only $K \times K$ inversion.

Computational Acceleration

Chapter 4 derived a correct and convergent Laplace-EM algorithm, but left its dominant per-iteration cost unaddressed. Each Newton step in E-step 4.2 requires forming and solving the $Q \times Q$ system (4.5) at cost $\mathcal{O}(I_j Q^2 + Q^3)$, in the meanwhile, each inner iteration of the M-step factor update (4.14)-(4.15) requires forming Σ^{-1} densely at cost $\mathcal{O}(Q^3)$. For large Q , these costs are not merely inconvenient but make the algorithm impractical. This chapter develops an exact acceleration method to remove this bottleneck by showing that the Poisson-Multinomial equivalence that already underlies the M-step can be redeployed via auxiliary-variable construction, which can remove the cross-category coupling from the E-step Hessian as well.

5.1 PME Auxiliary-Variable Construction

For a fixed group j , we introduce an auxiliary scalar ξ_i per observation $i \in \mathcal{I}_j$ and define the Poisson surrogate log-posterior

$$\ell_P(\lambda, \xi) = \sum_{i \in \mathcal{I}_j} \sum_{q=1}^Q \left[y_{iq} \eta_{iq}(\lambda) - \exp(\eta_{iq}(\lambda) + \xi_i) \right] - \frac{1}{2} \lambda^\top \Sigma^{-1} \lambda, \quad (5.1)$$

which treats the counts as independent Poisson draws with rate $\exp(\eta_{iq} + \xi_i)$ rather than as a multinomial draw. This is the same auxiliary-variable idea that underlies Proposition ??, but it is now applied to the E-step posterior instead of M-step likelihood.

5.2 Diagonal Low Rank Hessian & Woodbury Solve

With ξ fixed at $\xi^*(\lambda)$, we denote $d_q := \sum_{i \in \mathcal{I}_j} \exp(\eta_{iq}(\lambda) + \xi_i)$. The Hessian of $\ell_P(\cdot, \xi)$ with respect to λ is

$$\mathbf{H}_P(\lambda) = -\text{diag}(\mathbf{d}) - \Sigma^{-1}, \quad \mathbf{d} = (d_1, \dots, d_Q)^\top, \quad (5.2)$$

which is purely diagonal in its Poisson part. We expand Σ^{-1} by the Woodbury identity

$$\Sigma^{-1} = \sigma^{-2}\mathbf{I}_Q - \sigma^{-4}\mathbf{B}\mathbf{M}_K^{-1}\mathbf{B}^\top \text{ with } \mathbf{M}_K = \mathbf{I}_K + \sigma^{-2}\mathbf{B}^\top\mathbf{B}. \quad (5.3)$$

We write $\tilde{\mathbf{D}} = \text{diag}(\mathbf{d} + \sigma^{-2}\mathbf{1}_Q)$ for the combined diagonal term

$$-\mathbf{H}_P(\lambda) = \tilde{\mathbf{D}} - \sigma^{-4}\mathbf{B}\mathbf{M}_K^{-1}\mathbf{B}^\top, \quad (5.4)$$

it's a $Q \times Q$ matrix. Applying Woodbury identity once more and we obtain

$$(-\mathbf{H}_P(\lambda))^{-1} = \tilde{\mathbf{D}}^{-1} - \tilde{\mathbf{D}}^{-1}\mathbf{B}\Gamma^{-1}\mathbf{B}^\top\tilde{\mathbf{D}}^{-1} \text{ with } \Gamma := \mathbf{B}^\top\tilde{\mathbf{D}}^{-1}\mathbf{B} - \sigma^4\mathbf{M}_K. \quad (5.5)$$

Therefore, given the Cholesky factor $\mathbf{L}$ of $-\Gamma$, a Newton direction $\Delta\lambda = (-\mathbf{H}_P(\lambda))^{-1}\mathbf{g}$ for any gradient vector $\mathbf{g}$ is obtained as

$$\Delta\lambda = \tilde{\mathbf{D}}^{-1}\mathbf{g} + \tilde{\mathbf{D}}^{-1}\mathbf{B}(\mathbf{L}\mathbf{L}^\top)^{-1}\mathbf{B}^\top\tilde{\mathbf{D}}^{-1}\mathbf{g}, \quad (5.6)$$

computed by two matrix-vector products against $\mathbf{B}$ and one $K \times K$ triangular solve, after a one-time $\mathcal{O}(QK^2 + K^3)$ cost to form Γ and its Cholesky factor.

Proposition 1 (Negative definiteness and conditioning of Γ). If $\Gamma \prec 0$ and $\lambda_{\max}(\Gamma) \leq -\sigma^4$, then $-\Gamma \succ \sigma^4\mathbf{I}_K$ uniformly in λ , $\mathbf{B}$ and data.

Proof. Since $d_q \geq 0$ for each q , $\tilde{\mathbf{D}} = \text{diag}(\mathbf{d} + \sigma^{-2}\mathbf{1}_Q) \succeq \sigma^{-2}\mathbf{I}_Q$. Therefore, $\tilde{\mathbf{D}}^{-1} \preceq \sigma^2\mathbf{I}_Q$ and $\mathbf{B}^\top\tilde{\mathbf{D}}^{-1}\mathbf{B} \preceq \sigma^2\mathbf{B}^\top\mathbf{B}$. Rewrite the definition of Γ in (5.5) using the substitution $\mathbf{M}_K = \mathbf{I}_K + \sigma^{-2}\mathbf{B}^\top\mathbf{B}$, then

$$-\Gamma = \sigma^4\mathbf{M}_K - \mathbf{B}^\top\tilde{\mathbf{D}}^{-1}\mathbf{B} \succeq \sigma^4\mathbf{M}_K + \sigma^2\mathbf{B}^\top\mathbf{B} - \sigma^2\mathbf{B}^\top\mathbf{B} = \sigma^4\mathbf{I}_K.$$

□

Proposition 1 guarantees that the Cholesky factorization used in (5.6) is always numerically well behaved. Regardless of how large Q and $\mathbf{B}$ increase, the relevant $K \times K$ matrix never becomes ill-conditioned purely as an artefact of dimension, since its smallest eigenvalue is bounded below by the dimension-free quantity σ^4 .

5.3 Ascent Direction on True Posterior

The direction $\Delta\lambda$ of (5.6) is a Newton step for the surrogate objective $\ell_P(\cdot, \xi^*(\lambda))$ instead of true log-posterior ℓ .

Lemma 1 (Ascent direction). Let $\mathbf{g} = \nabla\ell(\lambda)$ and $\Delta\lambda = (-\mathbf{H}_P(\lambda))^{-1}\mathbf{g}$. If $\mathbf{g} \neq \mathbf{0}$, then $\mathbf{g}^\top\Delta\lambda > 0$.

Proof. Since $\Gamma \prec 0$ implies the Woodbury-expanded matrix in (5.5) is positive definite. Equivalently, (5.4) is positive definite because it is the negative Hessian of the strictly concave function $\ell_P(\cdot, \xi)$. Hence, $-\mathbf{H}_P(\lambda) = \tilde{\mathbf{D}} - \sigma^{-4}\mathbf{B}\mathbf{M}_K^{-1}\mathbf{B}^\top \succ 0$ by Proposition 1, so its inverse is also positive definite and $\mathbf{g}^\top\Delta\lambda = \mathbf{g}^\top(-\mathbf{H}_P(\lambda))^{-1}\mathbf{g} > 0$ for any $\mathbf{g} \neq \mathbf{0}$. □

Algorithm 1 E-step inner iteration: posterior-mode search for the group effect λ_j

Require: group data $\{y_{ij}\}_{i \in \mathcal{I}_j}$; current parameters $(\mu, \Phi, \mathbf{B}, \sigma^2)$; Armijo constants $c_1 \in (0, 1)$, $\rho \in (0, 1)$; tolerance $\epsilon > 0$

Ensure: posterior mode $\hat{\lambda}_j$ and curvature $-\mathbf{H}(\hat{\lambda}_j)$

- 1: initialize $\lambda^{(0)} \leftarrow \mathbf{0}$; $t \leftarrow 0$
- 2: **repeat**
- 3: re-profile the offset $\xi^{(t)} \leftarrow \xi(\lambda^{(t)})$
- 4: $\mathbf{g}^{(t)} \leftarrow \nabla \ell(\lambda^{(t)})$
- 5: form $-\mathbf{H}_P(\lambda^{(t)}) = \bar{\mathbf{D}}^{(t)} - \sigma^{-4} \mathbf{B} \mathbf{M}_K^{-1} \mathbf{B}^\top$
- 6: solve $(-\mathbf{H}_P(\lambda^{(t)})) \Delta \lambda^{(t)} = \mathbf{g}^{(t)}$
- 7: $s^{(t)} \leftarrow 1$
- 8: **while** $\ell(\lambda^{(t)} + s^{(t)} \Delta \lambda^{(t)}) < \ell(\lambda^{(t)}) + c_1 s^{(t)} \mathbf{g}^{(t)\top} \Delta \lambda^{(t)}$ **do**
- 9: $s^{(t)} \leftarrow \rho s^{(t)}$
- 10: **end while**
- 11: $\lambda^{(t+1)} \leftarrow \lambda^{(t)} + s^{(t)} \Delta \lambda^{(t)}$; $t \leftarrow t + 1$
- 12: **until** $\|\mathbf{g}^{(t)}\| \leq \epsilon$
- 13: $\hat{\lambda}_j \leftarrow \lambda^{(t)}$
- 14: **return** $\hat{\lambda}_j$ and $-\mathbf{H}(\hat{\lambda}_j)$

Let $\ell : \mathbb{R}^Q \rightarrow \mathbb{R}$ denote the true conditional log-posterior of the group effect λ , whose mode is defined by (4.6), and $\ell_P(\cdot, \xi)$ denotes the Poisson surrogate log-posterior with offset ξ . The Newton-type direction is

$$\Delta \lambda = (-\mathbf{H}_P(\lambda))^{-1} \mathbf{g}, \quad \mathbf{g} = \nabla \ell(\lambda), \quad (5.7)$$

where $\mathbf{H}_P(\lambda) = \nabla^2 \ell_P(\lambda, \xi)$ is the Hessian of the Poisson surrogate by the Woodbury expansion (5.4). We propose two regularity facts on which the analysis rests, and both of them are consequences of results already established for the model.

Assumption 1 (Curvature of true objective). Let $\ell \in C^2(\mathbb{R}^Q)$. On initial super-level set $\mathcal{L}_0 := \{\lambda \in \mathbb{R}^Q : \ell(\lambda) \geq \ell(\lambda^{(0)})\}$, there exist constants $0 < \mu_0 \leq L_0 < \infty$ with

$$\mu_0 \mathbf{I} \preceq -\nabla^2 \ell(\lambda) \preceq L_0 \mathbf{I}, \quad (5.8)$$

for all $\lambda \in \mathcal{L}_0$.

Assumption 1 holds for the model. The strong-concavity bound is supplied by the Gaussian random-effect prior alone. Since $\alpha \sim N(\mathbf{0}, \Sigma)$ with $\Sigma = \mathbf{B} \mathbf{B}^\top + \sigma^2 \mathbf{I}$, the prior contributes $\Sigma^{-1} \succeq (\theta_1 + \sigma^2)^{-1} \mathbf{I}$ to $-\nabla^2 \ell$, so $\mu_0 = (\theta_1 + \sigma^2)^{-1} > 0$ holds unconditionally. The smoothness bound is supplied by the multinomial log-likelihood, which has negative semi-definite curvature and whose operator norm is dominated by the total observed count $\|\nabla^2 \log p(\mathbf{y} \mid \lambda)\| \leq \sum_i y_{ij} < \infty$, so $L_0 = \lambda_{\max}(\Sigma^{-1}) + \sum_i y_{ij}$ is finite whenever the data are finite. Thus ℓ is μ_0 -strongly concave and L_0 -smooth by construction, and no additional sampling assumption is needed; the same fact guarantees that the maximizer $\hat{\lambda}$ exists and unique, and $\mathcal{L}_0$ is compact.

Assumption 2 (Uniform conditioning of the preconditioner). There exist constants $0 < m \leq M < \infty$ independent of t such that every preconditioner generated by Algorithm 1 satisfies,

$$m\mathbf{I} \preceq -\mathbf{H}_P(\boldsymbol{\lambda}^{(t)}) \preceq M\mathbf{I}, \quad (5.9)$$

for all iteration t .

Assumption 2 likewise follows from the model, subject to one mild and checkable condition on the data. The lower bound m is structural: by Proposition 4.2.1 the Woodbury-expanded matrix (5.3) is positive definite because $\boldsymbol{\Gamma} \prec \mathbf{0}$, uniformly over the parameter region of interest. The upper bound M requires that the diagonal working weights and the profiled offset stay bounded along the iterates; the working weights already satisfy $d_q \leq \sum_i M_i$, so the only quantity that could degrade the conditioning is the profiled offset ξ_{ij} , which enters through $\log y_{ij}$ and diverges only if a unit carries no counts at all. It therefore suffices that every included unit has at least one nonzero count,

$$y_{ij\cdot} = \sum_{q=1}^Q y_{ijq} \geq 1 \quad \text{for all } (i, j),$$

under which ξ_{ij} is finite and, being a continuous function of $\boldsymbol{\lambda}$ (Proposition 4.1.1) on the compact super-level set $\mathcal{L}_0$, keeps the spectrum of $-\mathbf{H}_P(\boldsymbol{\lambda}^{(t)})$ within a fixed interval $[m, M]$. This condition is innocuous in practice: empty units carry no information and are routinely removed during pre-processing of count and compositional data, and the same effect is achieved by the standard pseudo-count or truncation adjustments used for sparse counts. Assumption 2 thus holds for any dataset after the elementary pre-processing already standard in this setting.

Lemma 2 (Ascent Direction). Let $\mathbf{g} = \nabla \ell(\boldsymbol{\lambda})$ and $\Delta\boldsymbol{\lambda} = (-\mathbf{H}_P(\boldsymbol{\lambda}))^{-1} \mathbf{g}$ as in (5.7). If $\mathbf{g} \neq \mathbf{0}$, $\mathbf{g}^\top \Delta\boldsymbol{\lambda} > 0$. Moreover, the angle between $\Delta\boldsymbol{\lambda}$ and $\mathbf{g}$ is bounded away from a right angle,

$$\frac{\mathbf{g}^\top \Delta\boldsymbol{\lambda}}{\|\mathbf{g}\| \|\Delta\boldsymbol{\lambda}\|} \geq \frac{m}{M} > 0. \quad (5.10)$$

Proof. By Assumption 2, $-\mathbf{H}_P(\boldsymbol{\lambda})$ is symmetric positive definite, so its inverse is symmetric positive definite as well with spectrum contained in $[M^{-1}, m^{-1}]$. Hence, for any $\mathbf{g} \neq \mathbf{0}$,

$$\mathbf{g}^\top \Delta\boldsymbol{\lambda} = \mathbf{g}^\top (-\mathbf{H}_P(\boldsymbol{\lambda}))^{-1} \mathbf{g} \geq M^{-1} \|\mathbf{g}\|^2 > 0, \quad (5.11)$$

which is the first claim. For the angle, using $\mathbf{g}^\top (-\mathbf{H}_P(\boldsymbol{\lambda}))^{-1} \mathbf{g} \geq M^{-1} \|\mathbf{g}\|^2$ and $\|\Delta\boldsymbol{\lambda}\| = \|(-\mathbf{H}_P(\boldsymbol{\lambda}))^{-1} \mathbf{g}\| \leq m^{-1} \|\mathbf{g}\|$,

$$\frac{\mathbf{g}^\top \Delta\boldsymbol{\lambda}}{\|\mathbf{g}\| \|\Delta\boldsymbol{\lambda}\|} \geq \frac{M^{-1} \|\mathbf{g}\|^2}{m^{-1} \|\mathbf{g}\|^2} = \frac{m}{M} > 0. \quad (5.12)$$

□

Theorem 1 (Global Linear Convergence under Armijo Backtracking). Let $\lambda^{(0)} \in \mathbb{R}^Q$ be arbitrary and generate $\lambda^{(t+1)} = \lambda^{(t)} + s^{(t)} \Delta \lambda^{(t)}$, where $\Delta \lambda^{(t)}$ is computed from (5.6) at $\lambda^{(t)}$. For fixed Armijo parameters $c_1 \in (0, 1)$ and $\rho \in (0, 1)$, the step size $s^{(t)} = \rho^{k_t}$ is chosen with $k_t \in \{0, 1, 2, \dots\}$, which is the smallest integer satisfying the sufficient-increase condition,

$$\ell(\lambda^{(t)} + s^{(t)} \Delta \lambda^{(t)}) \geq \ell(\lambda^{(t)}) + c_1 s^{(t)} \mathbf{g}^{(t)\top} \Delta \lambda^{(t)}, \quad (5.13)$$

where $\mathbf{g}^{(t)} := \nabla \ell(\lambda^{(t)})$. Then the iterates converge to the unique maximizer $\hat{\lambda}$ of ℓ at a linear rate under Assumption 1-2. That is, there exists a constant $\kappa \in (0, 1]$ depending only on $(\mu_0, L_0, m, M, c_1, \rho)$ such that,

$$\ell(\hat{\lambda}) - \ell(\lambda^{(t)}) \leq (1 - \kappa)^t [\ell(\hat{\lambda}) - \ell(\lambda^{(0)})], \quad (5.14)$$

and,

$$\|\hat{\lambda} - \lambda^{(t)}\| \leq (1 - \kappa)^{\frac{t}{2}} \sqrt{\frac{2}{\mu_0} [\ell(\hat{\lambda}) - \ell(\lambda^{(0)})]}. \quad (5.15)$$

Proof. If $\mathbf{g}^{(t)} = \mathbf{0}$ for some t , then $\lambda^{(t)}$ is a stationary point of the strictly concave ℓ , hence $\lambda^{(t)} = \hat{\lambda}$ and the assertion holds trivially. Under generation assumption, we assume $\mathbf{g}^{(t)} \neq \mathbf{0}$ and proceed the proof in four steps.

Step 1. By Lemma 2, $\Delta \lambda^{(t)}$ is an ascent direction and $\mathbf{g}^{(t)\top} \Delta \lambda^{(t)} > 0$. Assumption 1 gives L_0 -Lipschitz continuity of $\nabla \ell$ on $\mathcal{L}_0$. Also, the property of monotonicity (proved below) of $\{\ell(\lambda^{(t)})\}$ keeps every iterate in $\mathcal{L}_0$. Thus, the quadratic lower bound,

$$\ell(\lambda^{(t)} + s \Delta \lambda^{(t)}) \geq \ell(\lambda^{(t)}) + s \mathbf{g}^{(t)\top} \Delta \lambda^{(t)} - \frac{L_0 s^2}{2} \|\Delta \lambda^{(t)}\|^2, \quad (5.16)$$

holds for all admissible s . Comparing (5.16) with (5.13), the Armijo condition is implied whenever

$$s \leq \frac{2(1 - c_1) \mathbf{g}^{(t)\top} \Delta \lambda^{(t)}}{L_0 \|\Delta \lambda^{(t)}\|^2}. \quad (5.17)$$

Since backtracking multiplies the trial step by ρ until acceptance and starts from 1, the accepted step satisfies,

$$s^{(t)} \geq \min \left\{ 1, \frac{2\rho(1 - c_1) \mathbf{g}^{(t)\top} \Delta \lambda^{(t)}}{L_0 \|\Delta \lambda^{(t)}\|^2} \right\}. \quad (5.18)$$

For the simplicity of the notation, we write $P^{(t)} = (-\mathbf{H}_P(\lambda))^{-1}$ with spectrum in $[M^{-1}, m^{-1}]$. From Lemma 2, we have,

$$\frac{\mathbf{g}^{(t)\top} \Delta \lambda^{(t)}}{\|\Delta \lambda^{(t)}\|^2} \geq \frac{m^2}{M}, \quad (5.19)$$

and hence,

$$s^{(t)} \geq s_{\min} := \min \left\{ 1, \frac{2\rho(1-c_1)m^2}{L_0M} \right\} > 0, \quad (5.20)$$

which is the bound independent of t . In particular, the line search terminates after finitely many backtracking steps.

Step 2. Combining the accepted Armijo inequality (5.13) with (5.20) and $\mathbf{g}^\top \Delta \lambda \geq M^{-1} \|\mathbf{g}\|^2$, we have,

$$\ell(\lambda^{(t+1)}) - \ell(\lambda^{(t)}) \geq c_1 s^{(t)} \mathbf{g}^{(t)\top} \Delta \lambda^{(t)} \geq \frac{c_1 s_{\min}}{M} \|\mathbf{g}^{(t)}\|^2. \quad (5.21)$$

We define the positive constant term $\gamma := \frac{c_1 s_{\min}}{M}$. Thus, $\{\ell(\lambda^{(t)})\}$ is non-decreasing and bounded above by $\ell(\hat{\lambda})$. It converges and every iterate lies in $\mathcal{L}_0$, which justify the use of Assumptions 1-2 and of (5.13) above.

Step 3. Assumption 1 yields the strong concavity. That is, for every $\lambda \in \mathcal{L}_0$, the quadratic upper bound satisfies that

$$\ell(\mathbf{y}) \leq \ell(\lambda)^\top (\mathbf{y} - \lambda) - \frac{\mu_0}{2} \|\mathbf{y} - \lambda\|^2, \quad (5.22)$$

which maximize the right-hand side over $\mathbf{y} \in \mathbb{R}^Q$. By Polyak-Lojasiewicz inequality, we have,

$$\ell(\hat{\lambda}) - \ell(\lambda) \leq \frac{1}{2\mu_0} \|\nabla \ell(\lambda)\|^2. \quad (5.23)$$

Let $\delta_t := \ell(\hat{\lambda}) - \ell(\lambda) \geq 0$. Subtracting (5.21) from (5.22) from $\ell(\lambda)$ and applying (5.23) at $\lambda^{(t)}$,

$$\delta_{t+1} = \delta_t - \left[\ell(\lambda^{(t+1)}) - \ell(\lambda^{(t)}) \right] \leq \delta_t - \gamma \|\mathbf{g}^{(t)}\|^2 \leq \delta_t - 2\gamma\mu_0\delta_t = (1 - \kappa)\delta_t, \quad (5.24)$$

where $\kappa := 2\gamma\mu_0 = \frac{2c_1 s_{\min}\mu_0}{M}$. Since $\delta_t \geq 0$, $\kappa \in (0, 1]$ and $\delta_t \leq (1 - \kappa)^t \delta_0$, which is the first stated bound and shows $\ell(\hat{\lambda}) \rightarrow \ell(\lambda)$.

Step 4. Strong concavity also gives $\ell(\hat{\lambda}) \leq \ell(\lambda) - \frac{\mu_0}{2} \|\hat{\lambda} - \lambda\|^2$. Hence,

$$\frac{\mu_0}{2} \|\hat{\lambda} - \lambda^{(t)}\|^2 \leq \delta_t \leq (1 - \kappa)^t \delta_0, \quad (5.25)$$

then,

$$\|\hat{\lambda} - \lambda^{(t)}\| \leq (1 - \kappa)^{\frac{t}{2}} \sqrt{\frac{2\delta_0}{\mu_0}} \rightarrow 0. \quad (5.26)$$

Therefore, $\lambda^{(t)} \rightarrow \hat{\lambda}$, which is the unique maximizer of ℓ . $\square$

Remark 1. The backtracking condition in Theorem 1 is on the true posterior ℓ instead of checking on the surrogate ℓ_p . It guarantees that no majorization error from

the Poisson approximation accumulates in the final iterate $\hat{\lambda}$, which is therefore exactly the mode defined by (4.6), to numerical precision, regardless of how the search direction that got there was computed.

5.4 Curvature Problem & Hybrid Algorithm

The posterior mode $\hat{\lambda}_j$ of the true multinomial log-posterior ℓ in E-step for each group j . Theorem 1 guarantees that the Woodbury-accelerated iteration reaches it. However, the Laplace approximation needs the curvature at that mode, and here the two Hessians that the algorithms manipulates part ways. We make the distinction precise, because it is both the source of Woodbury of the speed-up and the source of a small, quantifiable bias.

Recall the linear predictor (3.6) and the probability $\mathbf{p}_{ij} = \text{softmax}(\boldsymbol{\eta}_{ij}) \in \mathbb{R}^Q$ for categories. The group effect enters each category additively, where $\frac{\partial \eta_{ijq}}{\partial \lambda_{jr}} = \delta_{qr}$, so the score of the true multinomial log-posterior is given by,

$$\nabla \ell(\boldsymbol{\lambda}_j) = \sum_{i \in \mathcal{I}_j} (\mathbf{y}_{ij} - \mathbf{y}_{ij} \cdot \mathbf{p}_{ij}) - \boldsymbol{\Sigma}^{-1} \boldsymbol{\lambda}_j, \quad (5.27)$$

and the Hessian matrix is given by,

$$-\mathbf{H}_j(\boldsymbol{\lambda}_j) = \sum_{i \in \mathcal{I}_j} \mathbf{y}_{ij} \cdot \left(\text{diag}(\mathbf{p}_{ij}) - \mathbf{p}_{ij} \mathbf{p}_{ij}^\top \right) + \boldsymbol{\Sigma}^{-1}. \quad (5.28)$$

The prior precision $\boldsymbol{\Sigma}^{-1} = \left(\mathbf{B} \mathbf{B}^\top + \sigma^2 \mathbf{I}_Q \right)^{-1} = \sigma^{-2} \mathbf{I}_Q - \sigma^{-4} \mathbf{B} \mathbf{M}_K^{-1} \mathbf{B}^\top$ is diagonal-plus-low-rank-K by the Woodbury identity with $\mathbf{M}_K$ defined in (5.3). However, the likelihood of (5.28) carries the I_j outer products $-\mathbf{y}_{ij} \cdot \mathbf{p}_{ij} \mathbf{p}_{ij}^\top$. It couples all categories together and makes $-\mathbf{H}_j$ be a dense $Q \times Q$ matrix. The computational cost $\mathcal{O}(Q^3)$ prevents the direct Newton solve.

That's why we introduce the Poisson-Multinomial equivalence at the beginning in Chapter 3. The Poisson-Multinomial equivalence represents each count as an independent Poisson variate $y_{ijq} \sim \text{Poisson}(\exp(\eta_{ijq} + \xi_{ij}))$ with offset $\xi_{ij} = \log y_{ij} - \log \sum_q \exp(\eta_{ijq})$ re-profiled at the current $\boldsymbol{\lambda}_j$ so that $\exp(\eta_{ijq} + \xi_{ij}) = y_{ij} \cdot p_{ijq}$. The surrogate objective ℓ_P is the Poisson log-posterior with ξ_{ij} held fixed at this profiled value. The score of the surrogate Poisson is,

$$\nabla \ell_P(\boldsymbol{\lambda}_j) = \sum_{i \in \mathcal{I}_j} (\mathbf{y}_{ij} - \exp(\boldsymbol{\eta}_{ij} + \boldsymbol{\xi}_{ij})) - \boldsymbol{\Sigma}^{-1} \boldsymbol{\lambda}_j = \sum_{i \in \mathcal{I}_j} (\mathbf{y}_{ij} - \mathbf{y}_{ij} \cdot \mathbf{p}_{ij}) - \boldsymbol{\Sigma}^{-1} \boldsymbol{\lambda}_j, \quad (5.29)$$

coincides with the true multinomial score in (5.27). The surrogate and the target share the same stationary point, which is why the line search recovers the exact multinomial model. The Hessians differ. Holding ξ_{ij} fixed makes each Poisson coordinate a

function of its own λ_{jq} alone, so the cross-category second derivatives vanish and,

$$-\mathbf{H}_P(\lambda_j) = \text{diag} \left(\sum_{i \in \mathcal{I}_j} y_{ij} \cdot \mathbf{p}_{ij} \right) + \boldsymbol{\Sigma}^{-1} = \tilde{\mathbf{D}} - \sigma^{-4} \mathbf{B} \mathbf{M}_K^{-1} \mathbf{B}^\top, \quad (5.30)$$

where $\tilde{d}_q = \sum_{i \in \mathcal{I}_j} y_{ij} \cdot p_{ijq} + \sigma^{-2}$. Therefore, the surrogate negative Hessian is diagonal-plus-rank- K , which is the structure solved by Woodbury equality and gives the Newton direction $(-\mathbf{H}_P)^{-1} \nabla \ell$ and the log-determinant $\log |-\mathbf{H}_P|$ with cost $\mathcal{O}(QK^2 + K^3)$ instead of $\mathcal{O}(Q^3)$.

In the meanwhile, the Equation (5.28) shows two combination of Hessian matrix. The first term $\sum_{i \in \mathcal{I}_j} y_{ij} \cdot (\text{diag}(\mathbf{p}_{ij}) - \mathbf{p}_{ij} \mathbf{p}_{ij}^\top)$ is given by data itself, which is positive semi-definite and scales with the total $y_{ij\cdot}$. It carries more curvature and hence a tighter posterior data-driven precision. The second (prior) term $\boldsymbol{\Sigma}^{-1}$ is anisotropic. We do the eigen-decomposition,

$$\mathbf{B} \mathbf{B}^\top = \sum_k \theta_k \mathbf{u}_k \mathbf{u}_k^\top, \quad (5.31)$$

and,

$$\mathbf{P}_\perp = \mathbf{I}_Q - \sum_k \mathbf{u}_k \mathbf{u}_k^\top, \quad (5.32)$$

for the projection onto the $(Q - K)$ -dimensional null space of $\mathbf{B}^\top$,

$$\boldsymbol{\Sigma}^{-1} = \frac{1}{\sigma^2} \mathbf{P}_\perp + \sum_k \frac{1}{\theta_k + \sigma^2} \mathbf{u}_k \mathbf{u}_k^\top, \quad (5.33)$$

On the K loading directions, the prior contributes only $(\theta + \sigma^2)^{-1}$, which is negligible in the data-dominated pervasive regime $\theta_k \asymp Q$. On the $Q - K$ leftover directions orthogonal to the loadings, the prior contributes exactly σ^{-2} , which is the entire structural curvature our model places here. Thus, the idiosyncratic variance σ^2 is the parameter that governs posterior precision precisely in the directions diverges and the posterior collapses onto them.

Proposition 2. For every λ_j ,

$$-\mathbf{H}_j(\lambda) = -\mathbf{H}_P(\lambda) - \sum_{i \in \mathcal{I}_j} y_{ij} \cdot \mathbf{p}_{ij} \mathbf{p}_{ij}^\top. \quad (5.34)$$

The correction is positive semi-definite of rank at most $I_j = |\mathcal{I}_j|$, consequently $-\mathbf{H}_P(\lambda) \succeq -\mathbf{H}_j(\lambda) \succ 0$.

Proof. Subtract (5.30) from (5.28), the diagonal terms $\text{diag}(\sum_i y_{ij} \cdot \mathbf{p}_{ij})$ and the prior precision $\boldsymbol{\Sigma}^{-1}$ cancel, leaving the equality (5.34). Each summand $y_{ij} \cdot \mathbf{p}_{ij} \mathbf{p}_{ij}^\top$ is a non-negative scalar times an outer product, hence positive semi-definite of rank one. Their sum has rank at most I_j . Positive semi-definiteness of the correction gives

$-\mathbf{H}_P \succeq -\mathbf{H}_j$, and $-\mathbf{H}_j \succ 0$ because it is the negative Hessian of the strictly concave ℓ . $\square$

Therefore, the Laplace covariance of Poisson surrogate and true Multinomial satisfy the positive semi-definite order,

$$\hat{\mathbf{S}}_j^P := (-\mathbf{H}_P(\hat{\lambda}_j))^{-1} \preceq \hat{\mathbf{S}}_j, \quad (5.35)$$

That is, the Poisson surrogate systematically understates posterior uncertainty. Substituting $\hat{\mathbf{S}}_j^P$ for $\hat{\mathbf{S}}_j$ in Rubin-Thayer update of $(\mathbf{B}, \sigma)$ in (4.11) would introduce a systematic downward bias into $\hat{\sigma}^2$ and into $\mathcal{L}^{(t)}$.

The resolution is the hybrid strategy that gives the name of this section. The cheap Woodbury-accelerated surrogate Newton iteration is used to find the model $\hat{\lambda}_j$ and evaluate the true multinomial Hessian once to obtain $\hat{\mathbf{S}}_j$ at that mode.

Theorem 1 guarantees that the Woodbury-accelerated iteration converges to the correct mode $\hat{\lambda}$. It says nothing, however, about the curvature at that mode, and the curvature is what the Laplace approximation (4.8) actually needs

$$\hat{\mathbf{S}}_j = (-\mathbf{H}_j(\hat{\lambda}_j))^{-1}, \quad (5.36)$$

with $\mathbf{H}_j$ the Hessian of the true multinomial log-posterior ℓ , not of the Poisson surrogate ℓ_P .

Although $\hat{\mathbf{S}}_j$ costs $\mathcal{O}(Q^3)$, we prefer using the exact value instead of Poisson surrogate (except the cases with super large Q) because the calculation of $\hat{\Sigma}_{\text{obs}}$ and Rubin-Thayer update (4.14)-(4.15) are dependent of it. Any error in the curvature propagates into the parameter estimates. Proposition 2 shows that the surrogate Poisson over-states curvature by a positive-semi-definite amount, so the surrogate Laplace covariance $\hat{\mathbf{S}}_j^P$ in (5.35) under-states the posterior variance. Through (5.33) this bias falls hardest on the leftover directions that σ^2 alone controls, and it feeds a downward bias into σ^2 at the next M-step. Quantifying and removing this bias is the purpose of the ordering established next and of the hybrid strategy that follows.

5.5 Complexity Analysis

5.5.1 Numerical Equivalence of Two Implementations

Before quantifying the speed-up efficiency, we establish that it is not bought at the cost of accuracy. The Woodbury implementation and the dense reference compute the same quantities because the Woodbury identity being an exact algebraic equivalence rather than an approximation. We verify this directly at the three points where the two implementations differ in arithmetic but must agree in value.

Table 5.1: Numerical equivalence of the dense and Woodbury implementations ($K = 2$, single group). Errors are maximum absolute differences; the mode is compared by both correlation and maximum absolute difference.

Q	$\Sigma^{-1}v$ error	mode corr.	mode max. diff.	$\log \hat{\mathbf{S}}_j $ error
50	$1.1e^{-14}$	1.0000	$5.3e^{-8}$	[X]
100	$2.8e^{-14}$	1.0000	$4.6e^{-10}$	[X]
200	$9.6e^{-14}$	1.0000	$3.6e^{-10}$	[X]
500	$6.3e^{-14}$	1.0000	$5.0e^{-10}$	[X]
1000	$2.6e^{-13}$	1.0000	$2.4e^{-8}$	[X]

5.5.2 Numerical Gap of $\mathbf{S}$ and $\mathbf{S}^P$

The equivalence of Table 5.2 established that the two implementations agree on the posterior model. However, the mode is affected both by the output of E-step and Laplace curvature $\hat{\mathbf{S}}_j$. We have found in Proposition 2 and (5.35) that Poisson surrogate $\hat{\mathbf{S}}_j^P$ over-states the curvature and hence under-states the variance, which has been shown by the estimation of σ^2 in Table 5.2.

Table 5.2: Full-EM comparison of the dense reference against the Woodbury fitter with the exact and surrogate posterior curvatures ($Q = 30$, $K = 2$, $J = 20$, $I_j = 12$, $\sigma_{\text{true}}^2 = 0.30$). The mode-based quantities are shared. The two Woodbury variants differ only in $\hat{\mathbf{S}}_j$.

Quantity	Dense + $\mathbf{S}$	Woodbury + $\mathbf{S}$	Woodbury + $\mathbf{S}^P$
Final Laplace log-ev	-103349.10	-103349.08	-103379.49
$\hat{\sigma}^2$	0.2556	0.2557	0.2447
$d(\hat{\mathbf{B}}, \mathbf{B}_{\text{true}})$	0.5019	0.4956	0.4743
λ mode correlation	0.9514	0.9515	0.9515
log-ev non-decreasing	Yes	Yes	Yes
$ \log\text{-ev} - \text{dense} $	—	0.017	30.39
$ \hat{\sigma}^2 - \text{dense} $	—	0.0001	0.0109

5.5.3 Acceleration Efficiency

Across all Q , the prior-precision action and the analytic log-determinant agree with the dense computations to [X] (order 10^{-14} , i.e. floating-point round-off), and the two posterior modes coincide with correlation [X] and maximum absolute difference [X] (order 10^{-8} , the tolerance of the shared gradient stopping rule). The two implementations are thus numerically indistinguishable, and the timing comparisons that follow contrast two exact routes to identical estimates, differing only in computational cost.

Table 5.3: Per-operation asymptotic complexity: dense base vs. Woodbury. Q categories, K factors ($K \ll Q$), I_j within-group observations, P covariates.

Operation (per group / per inner iteration)	Dense	Woodbury	Speed-up
E-step Newton direction $(-\mathbf{H})^{-1}\mathbf{g}$	$O(Q^3)$	$O(QK^2 + K^3)$	$O(Q^2/K^2)$
Gradient / softmax / line-search ℓ	$O(I_j Q)$	$O(I_j Q)$	1
Posterior curvature $\hat{\mathbf{S}}_j$	$O(Q^3)$	$O(QK^2)$	$O(Q^2/K^2)$
$\log \hat{\mathbf{S}}_j $	$O(Q^3)$	$O(QK^2 + K^3)$	$O(Q^2/K^2)$
Rubin-Thayer $\mathbf{B}$ update (per RT iter)	$O(Q^3)$	$O(Q^2 K)$	$O(Q/K)$
$\log \boldsymbol{\Sigma} $	$O(Q^3)$	$O(K^3)$	$O(Q^3/K^3)$
Fixed-effect $\mu, \boldsymbol{\phi}$ (Q Poisson GLMs)	$O(Q g(I_j, P))$	$O(Q g(I_j, P))$	1

Table 5.3 isolates the source of the speed-up. Every operation whose dense cost is $O(Q^3)$ becomes $O(QK^2)$ or $O(Q^2 K)$ under the Woodbury factorisation, while the two operations that are *not* functions of the diagonal-plus-low-rank structure—the $O(I_j Q)$ objective/softmax evaluations and the Q independent Poisson GLMs of the fixed-effect step—are, correctly, left unchanged. The acceleration is therefore confined to exactly the linear-algebra that the PME-induced structure of §?? makes reducible, and to nothing else; this is why the per-operation speed-up columns scale as Q^2/K^2 for the E-step and Q/K for the Rubin–Thayer update, and not as a single global constant.

Neyman-Scott Problems Extension

The consistency result of Theorem ?? in Chapter 5 is reassuring but addresses the regime opposite to the one that motivates this thesis. There we fixed the category dimension Q and let the number of groups $J \rightarrow \infty$. In that classical settings, the loading matrix $\mathbf{B}$ is estimated consistently. However, the applications of Chapter 1 live in the regime $Q \rightarrow \infty$ with J fixed. This chapter shows that the two regimes differ not merely quantitatively but qualitatively. In the large- Q regime the estimator of the loading subspace ceases to be consistent, and the mechanism responsible is structural relative of the classical incidental-parameter problem.

6.1 Incidental Parameter Problem Transposed

Scott and Neyman [1948] exhibited a class of models in which the maximum likelihood estimator is inconsistent. In their canonical example, $y_{jk} \sim \mathcal{N}(\mu_j, \sigma^2)$ for j groups and k replicates per group, with n fixed and $J \rightarrow \infty$. Each group carries its own nuisance mean μ_j , so the number of parameters grows with the sample size. Profiling them out leaves a downward-biased estimator (ichi Amari [2016])

$$\hat{\sigma}^2 = \frac{n-1}{n} \sigma^2, \quad (6.1)$$

which never converges to the true value. Lancaster [2000] traced the many settings in which this pathology recurs.

Our model inverts this arithmetic. The nuisance parameters are the group random effects $\{\lambda_j\}_{j=1}^J$, whose number J is fixed. The structural parameter is the loading matrix $\mathbf{B} \in \mathbb{R}^{Q \times K}$, whose dimension QK diverges as Q goes to infinity. Different from the classical problem with many low-dimensional nuisance parameter corrupting a fixed target, we have a single high-dimensional target supported by a fixed number of replicates. We call this the transposed incidental-parameter problem.

The transposition preserves the essential difficulty. Consistent estimation of $\mathbf{B}$ requires the data information of each loading direction accumulate without bound. That information is supplied by independent realizations of the group effect, of which there are exactly J . Enlarging the within-group sample size of each group I_j sharpens the posterior of each individual λ_{ij} but adds no new independent repli-

cate of the latent factor. Enlarging Q adds new coordinates of $\mathbf{B}$ to be estimated rather than new information about the existing ones. Consequently, the effective sample size for the column space of $\mathbf{B}$ is J , not QJ . When J is fixed and $Q \rightarrow \infty$, the information per loading is bounded.

We separate this phenomenon from repairable bias. Conditional likelihood proposed by Liang [1987] and adjust likelihood proposed by McCullagh and Tibshirani [1990] try to modify the profile likelihood so that it may be expected to behave more like an honest likelihood. Their corrections succeed because the information lost to the nuisance parameters is recoverable. The loading-direction inconsistency studied here is of a different kind. It reflects an information-theoretic ceiling, but not a correctable distortion. With J fixed, there is only a finite amount of information about the QK -dimensional subspace. Our contribution is therefore to quantify the consistency, but not to repair it.

Two consequences of this ceiling organize the remainder of this chapter. The first is a direct analogue of the classical variance bias. The idiosyncratic variance estimator inherits a $\frac{J-K}{J}$ downward bias, which is the factor-analytic counterpart of the $\frac{n-1}{n}$ deflation in the Neyman-Scott normal example and vanishing only as $J \rightarrow \infty$. The second is the split in the asymptotic behavior of the estimator across its blocks. The fixed-effect and score directions governed by the Q independent Poisson subproblems of the Poisson-Multinomial equivalence in section 3.2

6.2 Fisher Information

This section constructs the Fisher information of the observed data for the loading matrix $\mathbf{B} \in \mathbb{R}^{Q \times K}$ and establishes the bound of the plateau. Two structural facts drive everything. First, the loading matrix $\mathbf{B}$ enters

Simulation Studies & Data Application

In this chapter, we lay out numerical results illustrating the computational and statistical efficiency of the algorithm proposed in Chapter 4 and Chapter 5. We present the results for parameter estimation and asymptotic inference respectively.

7.1 Base Model & Woodbury

Throughout Chapter 5, we focus on the high-dimensional EM algorithm in large-Q regime, which is illustrated in Algorithm ???. We empirically study two cases using our algorithm: large-J regime and large-Q regime motivated this thesis. Before comparing the consistency rates and efficient, we provide the statement proved by Figure ??? saying that the error between estimators given by base EM algorithm and Woodbury EM algorithm can be controlled near the machine error, which demonstrates that these two algorithms give the same estimation numerically.

Figure ??

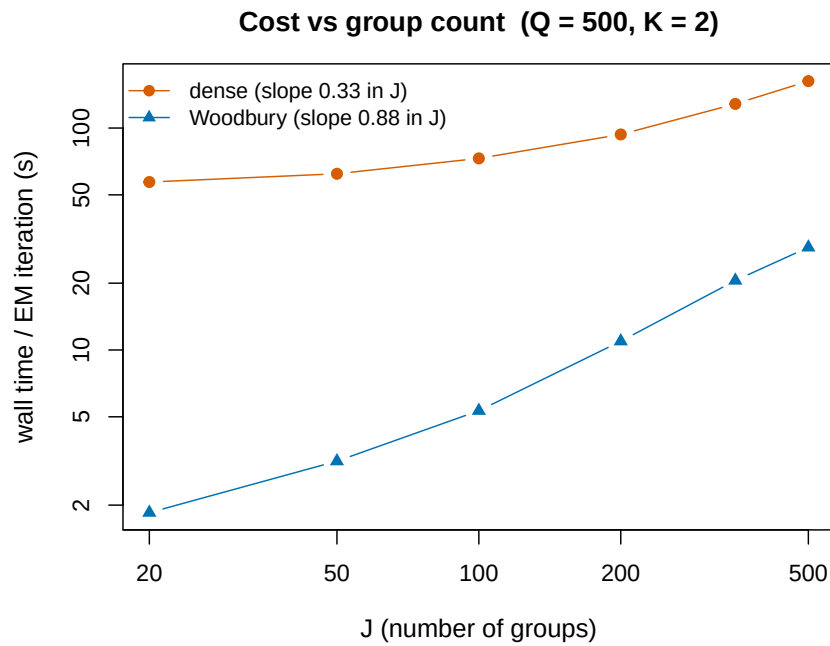


Figure 7.1: Time Cost Comparison.

The first result under large- J regime states that our Woodbury EM algorithm can reduce the cost of time when J increasing, with large and fixed Q . Before the test on real-data, we do the simulation on generated dataset.

Conclusion

8.1 Future Work

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